

Functional Annotation of DnaE2 (gene *dnaE2* / PP_3119; UniProt Q88I82)

Pseudomonas putida KT2440

1. Summary (Answer to the Research Question)

DnaE2 (Q88I82, locus PP_3119) is a second, non-essential catalytic subunit of DNA polymerase III belonging to the C-family of DNA polymerases (PolC/DnaE; EC 2.7.7.7, DNA-directed DNA polymerase). Its primary function is **error-prone, template-directed DNA synthesis across DNA lesions (translesion synthesis, TLS)** as the catalytic engine of the bacterial "**mutasome**." It catalyses the addition of deoxyribonucleotides to a primer 3'-OH using a damaged DNA template as instruction, producing DNA synthesis past bulky/blocking lesions at the cost of introducing point mutations. DnaE2 is **induced by DNA damage as part of the RecA/LexA-controlled SOS response** and acts together with two accessory proteins encoded in the same cassette — **ImuA'** (a RecA paralog) and **ImuB** (a catalytically dead Y-family polymerase that tethers the complex to the β sliding clamp). The gene product functions **in the cytoplasm, at the DNA replication fork**, where it is recruited to stalled replisomes. Physiologically it mediates **damage-induced (SOS) mutagenesis**, contributing to survival of genotoxic stress and to the evolution of adaptive mutations, including antibiotic/drug resistance.

Identity check (passed). The gene symbol, C-family (PolC/DnaE) assignment, the "DnaE2" subfamily InterPro signatures (IPR023073 DnaE2; IPR004805 DnaE2/DnaE/PolC; IPR011708 DNA_pol3_alpha NTPase; IPR040982 DNA_pol3_finger; IPR029460 DNAPol_HHH), and the EC 2.7.7.7 activity all correspond precisely to the *bona fide* error-prone DnaE2 polymerase family. The literature cited below concerns this same protein family in closely related and model bacteria; the *P. putida* ortholog PP_3119 is annotated by orthology (HAMAP rule MF_01902).

2. Background: what "a second DnaE" means

Most bacteria replicate their chromosome with a single, essential replicative polymerase DnaE (the α catalytic subunit of DNA Pol III). A subset of bacteria — including *Pseudomonas*, *Caulobacter*, *Mycobacterium*, and many other Proteobacteria and Actinobacteria — carry a **second, paralogous *dnaE* gene, *dnaE2***. The existence of these "extra" replicative-type polymerases was long an "unresolved enigma," resolved by the demonstration that one copy participates in **error-prone DNA repair synthesis** rather than in bulk genome replication (Boshoff et al. 2003, Cell,

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Importantly, DnaE2 substitutes functionally for the Y-family **UmuDC/Pol V** mutagenic system, which is **absent** in these organisms (Galhardo et al. 2005,

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P. putida KT2440 lacks *umuDC*, so DnaE2 provides the RecA-dependent branch of induced mutagenesis.

3. Primary Molecular Function (the catalyzed reaction and substrate)

- **Reaction:** DNA-templated (DNA-directed) DNA polymerization — nucleotidyl transfer: $(DNA)_n + dNTP \rightarrow (DNA)_{n+1} + diphosphate$ (EC 2.7.7.7).
- **Substrate specificity:** a **primer-template DNA duplex** with a free 3'-OH primer terminus, plus the four **dNTPs** and the catalytic divalent metal ions (Mg^{2+}). Unlike the high-fidelity replicative DnaE, DnaE2 is **error-prone** and, as part of the mutasome, is able to extend primers opposite **damaged/distorting template bases** (e.g., lesions produced by UV or by cross-linking agents such as mitomycin C). This lesion-bypass capacity is the defining, functionally important activity (Galhardo et al. 2005,

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Boshoff et al. 2003,

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- **Domain basis:** the PolIII α -type NTPase/polymerase domain (IPR011708) provides the catalytic center; the "polymerase-III finger" (IPR040982) and helix-hairpin-helix (HHH, IPR029460) domains support DNA binding — architecture shared with the replicative DnaE but tuned toward lower fidelity.

DnaE2 is **not** a general-purpose replicase: knockouts are viable and do not impair normal growth; the phenotype appears specifically after genotoxic stress (reduced survival and greatly reduced induced mutation frequency) (Boshoff 2003,

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Galhardo 2005,

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4. The Functional Unit: the ImuA'-ImuB-DnaE2 "mutasome"

DnaE2 does **not** act alone. In *P. putida* (as in *Caulobacter* and *Mycobacterium*) the *dnaE2* gene is the third gene of the **imuA-imuB-dnaE2 cassette**, and **all three products are required** for error-prone lesion processing (Galhardo et al. 2005,

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Recent biochemical/genetic work defines the roles of the partners:

- **ImuB** — a **catalytically inactive Y-family DNA polymerase**. It cannot itself synthesize DNA but carries a **β -clamp-binding motif** and "is predicted to enable mutasome function via its interaction with the β clamp" (Santos et al. 2025,

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ImuB thus acts as a **non-catalytic scaffold/adaptor** that tethers DnaE2 to the DNA Pol III sliding clamp at the replication fork.

- **ImuA'** — a **RecA homolog** of accessory (non-recombinase) function. ImuB carries a **RecA-N-terminus (RecA-NT) homology motif** that mediates the ImuB-ImuA' interaction; disrupting this motif (e.g., hydrophobic residues L378/V383) abolishes the ImuA'-ImuB interaction, and this interaction is **essential for DNA damage-induced mutagenesis**

(Santos et al. 2023,

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Santos et al. 2025,

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Decisive genetic-biochemical dissection (Warner et al. 2010,
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In *M. tuberculosis*: (i) *imuA'* and *imuB* are each **individually essential** for induced mutagenesis and damage tolerance; (ii) yeast two-hybrid shows **ImuB interacts simultaneously with ImuA', with DnaE2, and with the β -clamp** — i.e., ImuB is the central adaptor; (iii) **disrupting the ImuB- β -clamp interaction reduces induced mutagenesis and damage tolerance, phenocopying the deletion mutants** — establishing replisome recruitment as essential; (iv) ImuB retains Y-family fold but **lacks the conserved active-site residues** needed for catalysis; and, most importantly, (v) **replacing DnaE2's own catalytic residues reproduces the *dnaE2* deletion phenotype, directly implying that the DnaE2 α -subunit is the catalytic polymerase of mutagenic lesion bypass.** Gessner et al. 2023

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further dissect mutasome composition and recruitment.

Model: ImuB, docked on the β -clamp at a stalled fork, recruits ImuA' (via RecA-NT) and DnaE2. DnaE2 supplies the catalytic polymerase activity that inserts nucleotides opposite the lesion and/or extends the primer past it. This modular design explains how a C-family (normally replicative) polymerase performs specialized translesion synthesis only when assembled into the mutasome.

5. Subcellular Localization

The gene product functions in the **cytoplasm**, specifically **at the chromosomal DNA / the replication fork**. It is a **DNA-binding enzyme** recruited to stalled replisomes through the ImuB- β -clamp interaction (Santos et al. 2025,

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There is no signal peptide or transmembrane region expected for a C-family polymerase; its

site of action is wherever replication is blocked by template damage. Single-molecule/imaging studies of bacterial TLS confirm that lesion-bypass polymerases act at or behind the replication fork, either co-replicationally or on post-replicative gaps (Joseph & Badrinarayanan 2020,

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6. Pathway / Regulatory Context: the SOS (DNA-damage) response

- DnaE2/*imuAB dnaE2* is a member of the **RecA/LexA-controlled SOS regulon**. The operon is **damage-inducible in a RecA-dependent manner**, and induction requires DNA-damaging signals (UV, mitomycin C, and other genotoxins) (Galhardo et al. 2005,

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In *M. tuberculosis*, *dnaE2* is upregulated by several DNA-damaging agents and during infection (Boshoff 2003,

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- Mechanistically, DNA damage generates ssDNA, RecA forms a nucleoprotein filament, LexA autocleaves, and SOS genes (including *imuA–imuB–dnaE2*) are de-repressed — the canonical SOS pathway that operates in *P. putida*.
- **Functional output of the pathway:** conversion of otherwise lethal, replication-blocking lesions into survivable (but mutagenic) events — i.e., **DNA damage tolerance via error-prone translesion synthesis**. This is a *distinct* pathway from the *RecA-independent*, stationary-phase mutagenesis driven by Pol IV/DinB in *P. putida* (Tegova et al. 2004,

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which underscores that DnaE2 specifically provides the **RecA-dependent, SOS-inducible** mutagenic branch.

7. Physiological / Evolutionary Significance

- **Survival of genotoxic stress:** loss of DnaE2 reduces survival after UV irradiation (Boshoff 2003,

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- **Induced mutagenesis / adaptive evolution:** DnaE2 is responsible for most UV- and mitomycin-C-induced mutations in *Caulobacter*, including unusual substitution spectra (e.g., G:C→C:G transversions) (Galhardo 2005,

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- **Drug/antibiotic resistance:** by generating point mutations under stress, DnaE2 can directly contribute to the emergence of drug resistance in vivo (Boshoff 2003,

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— a clinically relevant consequence of the same TLS activity.

Genus-level (*Pseudomonas*) evidence. The closest well-studied relative, *Pseudomonas aeruginosa*, provides direct functional support: the **damage-inducible SOS-response *imuBC* gene products contribute (together with *dinB*/Pol IV and RpoS) to the generation of ciprofloxacin-resistance mutations** (Fahey et al. 2023,

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This demonstrates that the *imuAB dnaE2* module is an active, SOS-inducible mutagenesis system within the *Pseudomonas* genus and reinforces the annotation-by-orthology of *P. putida* PP_3119 as the catalytic translesion polymerase of that module.

For *P. putida* KT2440 specifically, these roles are inferred by **strong orthology** and conserved operon structure rather than by dedicated single-gene knockout studies; *P. putida* mutagenesis research has more often focused on Pol IV/DinB (RecA-independent stationary-phase mutagenesis,

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leaving a defined niche for DnaE2 as the RecA/SOS-dependent mutagenic polymerase.

8. Supported vs. Refuted Hypotheses

Supported - DnaE2 is a C-family (PolC/DnaE) DNA polymerase catalysing DNA-directed DNA synthesis (EC 2.7.7.7). (*family/domain + literature*) - Its specialized function is **error-prone translesion synthesis / damage-induced mutagenesis**, not bulk replication.

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- It operates within the **ImuA'-ImuB-DnaE2 mutasome**, tethered to the **β -clamp** at the replication fork.

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- **DnaE2 is the direct catalytic polymerase** for lesion bypass: mutating its catalytic residues phenocopies deletion, while its partner ImuB is catalytically dead.

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- It is **SOS/RecA-regulated** and induced by DNA damage.

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- Localization: **cytoplasmic, DNA/replication-fork associated.**

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Refuted / excluded - DnaE2 is the essential replicative replicase — **refuted** (knockouts viable; phenotype only under damage). - DnaE2 mediates the RecA-independent stationary-phase mutagenesis of *P. putida* — that role belongs to Pol IV/DinB (

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DnaE2 provides the RecA-dependent branch.

9. Limitations & Future Directions

- **Direct experimental evidence in *P. putida* KT2440 (PP_3119) is limited;** the functional annotation rests primarily on (i) unambiguous C-family/DnaE2 domain and family signatures, (ii) HAMAP orthology rule MF_01902, and (iii) detailed genetics/biochemistry of orthologs in *Caulobacter crescentus* and *Mycobacterium tuberculosis*.
- The precise *P. putida* mutational spectrum, the exact ImuA/ImuB residues, and in vivo fork localization for PP_3119 have not been directly measured.
- **Future work:** clean *dnaE2* / *imuB* deletions in *P. putida* KT2440 with UV/MMC survival and rifampicin-resistance mutagenesis assays; confirmation of a functional LexA/SOS box upstream of *imuA*; reconstitution of the *P. putida* mutasome to measure lesion-bypass fidelity and β -clamp dependence.

10. Key References

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